

Gauge Structure and Color Confinement in a Deterministic Brane-Lattice Substrate (Paper III)

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Abstract

We develop the gauge and color consistency bridges for the deterministic 4D brane-lattice substrate of Paper I. The gauge bridge shows that adiabatic transport of a band-isolated wavepacket's polarization eigenbundle induces a Berry connection $a_\mu = i\langle u | \partial_\mu u \rangle$ that is formally the electromagnetic four-potential A_μ , with Berry curvature $f_{\mu\nu}$ as the field tensor $F_{\mu\nu}$ and the homogeneous Maxwell equations following as the Bianchi identity $\partial_{[\lambda} F_{\mu\nu]} = 0$. The gauge structure does not live in the Brillouin zone: the dynamical matrix is real symmetric and carries identically zero Berry curvature there; we verify this numerically at every α . The physically meaningful gauge connection lives over physical spacetime (x, t) , where the $U(1)$ phase is rotation of the carrier envelope along the timelike direction. For a near-degenerate lateral triplet the Wilczek–Zee connection $\mathcal{A}_\mu \in \mathfrak{u}(3)$ generalizes the Berry connection to a non-Abelian gauge potential. The color bridge decomposes the axis-aligned triplet as $\mathfrak{u}(3) = \mathfrak{u}(1) \oplus \mathfrak{su}(3)$, identifies the $U(1)$ trace sector with electromagnetism and the traceless generators with color, and establishes kinematic confinement: no linear propagation direction simultaneously satisfies coherence (degenerate branches) and color-activity (non-degenerate branches with defined holonomy), so color charge cannot be carried to infinity by a free linear excitation. The falsifiable prediction is the off-[111] fibre-holonomy ratio $R(\alpha_2)/R(\alpha_1) = [(3 - 2\alpha_2)/\alpha_2]/[(3 - 2\alpha_1)/\alpha_1]$, equal to 0.3077 for $(\alpha_1, \alpha_2) = (0.2, 0.5)$.

Keywords: Berry connection; Wilczek–Zee connection; gauge emergence; color confinement; kinematic confinement; $U(3)$; $SU(3)$; brane lattice; band isolation; semilocal vortex; Skyrmion

1. Introduction

This is the third paper in a series developing a deterministic brane-lattice substrate hypothesis. Paper I established the substrate model and exported the linearized wave-branch structure. Paper II developed the Lorentz and gravity bridges. The present paper develops two closely coupled bridges: gauge structure and color confinement.

The gauge bridge. Each wavepacket carries a local polarization state, defined as a normalized eigenvector of the linearized carrier operator about the slowly varying background at the wavepacket's location. As the wavepacket is transported through regions of varying background, the eigenvector is parallel-transported along the connection it defines. This is the Berry connection (Section 3): for a single isolated mode, $a_\mu = i\langle u | \partial_\mu u \rangle$ is formally the electromagnetic four-potential A_μ , its curvature $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$ is the field tensor $F_{\mu\nu}$, and the homogeneous Maxwell equations follow as the Bianchi identity $F = dA$. No new

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assumption is added: gauge structure is not a coincidence but the canonical outcome of transporting a polarization eigenbundle.

A key point developed in Section 3.5: the gauge connection does *not* live in the Brillouin zone. The dynamical matrix $D(\mathbf{k})$ is real symmetric and carries identically zero Berry curvature over the BZ at every α ; we have verified this numerically. The physically relevant gauge connection lives over physical spacetime (x, t) , where the $U(1)$ phase is the rotation of the carrier field as one advances along the timelike direction.

The color bridge. The cubic substrate provides a natural three-dimensional polarization sector: the axis-aligned lateral triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top \in \mathbb{C}^3$. For a near-degenerate triplet (small α), adiabatic transport induces a Wilczek–Zee connection $\mathcal{A}_\mu \in \mathfrak{u}(3)$. Decomposing $\mathfrak{u}(3) = \mathfrak{u}(1) \oplus \mathfrak{su}(3)$ identifies the $U(1)$ trace sector with electromagnetism and the eight traceless generators with a color-like internal degree of freedom. The color bridge (Section 4) then establishes a structural fact: *kinematic confinement*. No free linear propagation direction simultaneously satisfies coherence (degenerate branches) and color-activity (non-degenerate branches with defined internal holonomy), so color charge cannot be carried to infinity by any free linear excitation. Color is structurally confined to nonlinear bound states — without invoking a separate QCD action.

Why gauge and color are paired. The two bridges share a common foundation: per-wavepacket band isolation of the lateral triplet. The gauge bridge extracts the Abelian rank-1 sector; the color bridge extracts the non-Abelian triplet sector. Both structures emerge from the same eigenbundle transport, and the $U(3)$ decomposition that distinguishes them is fixed by the cubic lattice geometry, not introduced by hand. Pairing them in one paper makes this shared derivation explicit.

Paper organization. Section 2 lists the interface equations imported from Paper I. Section 3 develops the gauge bridge: per-wavepacket isolation, Berry and Wilczek–Zee connections, the EM identification, where the gauge structure lives, and the spinorial holonomy. Section 4 develops the color bridge: axis-triplet degeneracy, $U(3)$ decomposition, topological sectors, kinematic confinement, and the numerical targets.

2. Imported interfaces from Paper I

This paper imports the following results from Paper I without re-derivation. All equation and section labels without an explicit paper prefix refer to Paper I.

Substrate model and equations of motion. The 4D brane embedding, induced metric g_{ij} (Equation (10)), constitutive law (Section 3.5), and equations of motion (Equation (26)).

Linearized branch structure. The long-wavelength dispersion $\omega_b^2(\mathbf{k}) \approx c_b^2 |\mathbf{k}|^2$ (Equation (32)), branch speeds $c_T^2 = (1 - \alpha)k_s a^2 / m$ and $c_L^2 = k_s a^2 / m$ (Equation (17)).

Axis-aligned triplet structure. The three spacelike embedding components (X^1, X^2, X^3) on the 6-neighbor cubic lattice define an axis-aligned triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top \in \mathbb{C}^3$, with α -driven degeneracy described in Section 4 of Paper I.

Geometric nonlinearity. The constitutive quartic coefficient $\propto k_s \alpha / a$ from Equation (20), which sets the energetic coupling between the $U(1)$ and $SU(3)$ sectors at finite amplitude.

Bell interpretive stance. Retrocausal worldtube reading from Section 5 of Paper I, inherited as a fixed commitment.

3. Gauge bridge: Berry/Wilczek–Zee connections as gauge structure

3.1. Per-wavepacket band isolation and polarization transport

The complex-envelope description is applied per band-isolated excitation, not to a globally coherent carrier sector spanning macroscopic scales. Each wavepacket is treated as narrowband around its own local carrier angular frequency ω_0 , determined by the linearized carrier operator about the slowly varying background at the wavepacket's location. A band-isolated wavepacket can be written locally as

$$\mathbf{u}(x, t) \approx \text{Re}\left\{a(x, t) \epsilon(x, t) e^{i\theta(x, t)}\right\}, \quad \theta(x, t) = \phi(x, t) - \omega_0 t, \quad (1)$$

where $a(x, t) \geq 0$ is the slowly varying envelope amplitude, $\epsilon(x, t) \in \mathbb{C}^N$ is a normalized polarization state ($\epsilon^\dagger \epsilon = 1$), and ω_0 is the wavepacket's local carrier angular frequency.

A central point is that polarization transport is not unique: the same physical state can be represented in many local phase conventions. Geometric phase formalizes this redundancy as a connection on a bundle of polarization states over the spacetime manifold parameterizing the wavepacket's trajectory.

3.2. Rank-1 Berry connection and electromagnetic identification

For a single isolated polarization mode, represent the local state by a normalized vector $|u(x)\rangle \in \mathbb{C}^N$ that depends smoothly on spacetime position x^μ (with $x^\mu = (ct, x^1, x^2, x^3)$ built from the intrinsic coordinates and a chosen branch speed c). Define the Berry connection

$$a_\mu(x) := i \langle u(x) | \partial_\mu u(x) \rangle. \quad (2)$$

Under a local phase choice $|u(x)\rangle \mapsto e^{-i\chi(x)} |u(x)\rangle$,

$$a_\mu \mapsto a_\mu + \partial_\mu \chi, \quad (3)$$

so only gauge-invariant quantities carry physical meaning. The Berry curvature is

$$f_{\mu\nu} := \partial_\mu a_\nu - \partial_\nu a_\mu, \quad (4)$$

and the holonomy around a closed loop Γ is

$$\gamma_\Gamma = \oint_\Gamma a_\mu dx^\mu = \int_{\Sigma: \partial\Sigma=\Gamma} f_{\mu\nu} d\Sigma^{\mu\nu}. \quad (5)$$

Electromagnetic reading: four-potential and field tensor. In the $U(1)$ trace sector (Section 4.2) the Berry connection (2) is the electromagnetic four-potential, $A_\mu \equiv a_\mu$, and its curvature (4) is the field tensor $F_{\mu\nu} \equiv f_{\mu\nu}$, with $E_i = F_{0i}$ (time–space) and $B_i = \frac{1}{2}\epsilon_{ijk}F_{jk}$ (space–space). Two consequences are automatic, requiring no field dynamics:

1. the gauge freedom (3) is exactly the unobservable choice of carrier phase reference, and
2. the homogeneous Maxwell equations are the Bianchi identity $\partial_{[\lambda} F_{\mu\nu]} = 0$, which holds identically because $F = dA$.

A nonzero field ($F \neq 0$) requires the phase to be non-integrable: the polarization state $|u\rangle$ must genuinely rotate within $\mathbb{C}^N$ ($N \geq 2$), not merely carry an overall phase, since a pure gradient $A_\mu = \partial_\mu \chi$ is pure gauge. What this kinematics does *not* fix is field *dynamics*: the inhomogeneous equations $d\star F = J$ hold only if the substrate furnishes a Maxwell action $-\frac{1}{4} \int F \wedge \star F$, a separate open derivation.

3.3. Eigenbundle picture and adiabaticity requirement

The connection (2) is not attached to an arbitrary amplitude/phase split. The physical choice is $|u(x)\rangle$ as a locally defined eigenvector of the linearized carrier operator about the slowly varying background at x . As x varies, these eigenvectors form an eigenbundle, and the Berry/Wilczek–Zee connection is the natural connection on that bundle — the same eigenbundle picture that underlies Berry-phase effects in crystalline electronic systems [1], here applied to the polarization eigenbundle of the mechanical carrier.

A clean rank-1 ($U(1)$) reduction requires dynamics to remain in a single isolated band. If $\Delta\omega_{\text{gap}}(x)$ is the local gap to the nearest competing band and $\Omega_{\text{mix}}(x)$ is the non-adiabatic mode-conversion rate induced by background gradients, the adiabatic condition

$$\Omega_{\text{mix}}(x) \ll \Delta\omega_{\text{gap}}(x) \quad (6)$$

must hold on the envelope scale. When (6) fails, the correct reduction retains the relevant multi-dimensional subspace (the Wilczek–Zee description). The prestress parameter α and wavelength regime jointly control branch separation and therefore the validity of the Abelian gauge description.

3.4. Wilczek–Zee connection for the degenerate triplet

For an n -dimensional near-degenerate subspace, choose an orthonormal frame $\mathbf{E}(x) \in \mathbb{C}^{N \times n}$ with columns $|u_a(x)\rangle$ spanning the subspace ($\mathbf{E}^\dagger \mathbf{E} = I_n$). The Wilczek–Zee connection is

$$\mathcal{A}_\mu(x) := i \mathbf{E}(x)^\dagger \partial_\mu \mathbf{E}(x) \in \mathfrak{u}(n). \quad (7)$$

Under $\mathbf{E} \mapsto \mathbf{E}U(x)$ with $U(x) \in U(n)$,

$$\mathcal{A}_\mu \mapsto U^\dagger \mathcal{A}_\mu U + i U^\dagger \partial_\mu U, \quad (8)$$

and the non-Abelian curvature is

$$\mathcal{F}_{\mu\nu} := \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - i[\mathcal{A}_\mu, \mathcal{A}_\nu]. \quad (9)$$

For $n = 3$ (the lateral triplet, Section 4), $\mathcal{A}_\mu \in \mathfrak{u}(3)$ decomposes into a $U(1)$ trace part and a traceless $SU(3)$ part, with the commutator term in (9) carrying the self-coupling of the non-Abelian sector.

Gauge-covariant envelope dynamics. Projecting the full substrate equations of motion onto the retained n -dimensional subspace produces a gauge-covariant derivative acting on the slow envelope $\Psi \in \mathbb{C}^n$:

$$\mathcal{D}_\mu \Psi = (\partial_\mu + i \mathcal{A}_\mu) \Psi, \quad (10)$$

which transforms covariantly under local frame rotations. The coarse-grained slow dynamics then admits a gauge-invariant effective action

$$S_{\text{eff}}[\Psi, \mathcal{A}] = \int d^4x \left(\Psi^\dagger \mathcal{K}(\mathcal{D}) \Psi - \kappa \text{tr}(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}) + \dots \right), \quad (11)$$

where $\mathcal{K}(\mathcal{D})$ is determined by the linearized carrier band and κ is an emergent curvature stiffness.

3.5. Where the gauge structure lives: spacetime, not the Brillouin zone

The gauge connection does *not* live in the Brillouin zone. The dynamical matrix $D(\mathbf{k})$ is the Hessian of a real-valued spring energy: real and symmetric for every $\mathbf{k}$. Real symmetric matrices admit real eigenvectors, and a real eigenvector bundle over any simply-

connected BZ region is topologically trivial. Every plaquette holonomy reduces to the identity and every Berry/Wilczek–Zee curvature vanishes identically. We have verified this numerically across a full α sweep over the entire (k_x, k_y) plane: holonomy is identity to machine precision at every plaquette and every α .

The physical gauge connection instead lives over spacetime (x, t) . It is the holonomy of the complex carrier envelope $\Psi \in \mathbb{C}^3$ transported around loops in real space and time, with the $U(1)$ trace read as the electromagnetic sector. The Brillouin zone and Bloch’s theorem enter only to supply the carrier band and to justify the narrowband isolation of a wavepacket about its carrier $\mathbf{k}_0$; they are not the base space of the gauge connection.

The reason the base space is (x, t) and not a spatial slice is that the complex structure itself originates in the timelike direction: the $U(1)$ phase is the rotation of the real carrier field as one advances along the time axis (Section 5 of Paper I). A single spacelike slice is a real snapshot and carries no phase; two adjacent time slices are the minimum data that fix the $U(1)$.

3.6. Spinorial holonomy and the 4π aspect

For an $n = 2$ subspace (a two-level polarization bundle), holonomy lives in a double cover of $SO(3)$. A closed loop in physical space that induces a 2π rotation of an internal polarization frame can therefore produce a sign change of the transported state, while a 4π loop returns it to itself. This spinorial 4π aspect is attributed to Berry/Wilczek–Zee holonomy of the internal polarization bundle. Its connection to physical spin- $\frac{1}{2}$ phenomenology is part of the matter bridge (Paper IV).

3.7. Gauge bridge open problems

1. **Carrier-mode propagation law.** The gauge connection structure $(A_\mu, F_{\mu\nu}$, covariant derivative) is established by the bridge. The outstanding task is to derive that free fluctuations of the Berry/Wilczek–Zee connection propagate as massless modes with Maxwell/Yang–Mills dispersion: the carrier-excitation spectrum in the appropriate limit should reproduce the massless spin-1 photon/gluon propagation law.
2. **Inhomogeneous Maxwell equations.** Show that the substrate furnishes the Maxwell action $-\frac{1}{4} \int F \wedge \star F$ (with its emergent Hodge star), so that the inhomogeneous equations $d\star F = J$ hold with the correct coupling.
3. **Adiabaticity in the nonlinear regime.** Verify that the adiabatic condition (6) holds in the localized-mode parameter regime and quantify the non-adiabatic correction to the holonomy.

4. Color bridge: axis-triplet $U(3)$ structure and kinematic confinement

4.1. Axis-aligned triplet and prestress-driven degeneracy

On the 6-neighbor axial-only cubic lattice it is natural to organize the per-wavepacket complex envelope as a three-component triplet $\Psi = (\psi_x, \psi_y, \psi_z)^\top \in \mathbb{C}^3$ aligned with the lattice axes. The prestress parameter α controls the eigenvalue spread of the 3-channel Christoffel matrix $D(k)$ without introducing off-diagonal coupling between axis components.

Non-degenerate limit ($\alpha \rightarrow 1$, no prestress). The lattice energy is purely Hookean longitudinal; $D(k)$ has maximally split eigenvalues. The three axis channels are dynamically decoupled, each supporting an independent rank-1 Berry connection $a_\mu^{(x)}, a_\mu^{(y)}, a_\mu^{(z)}$. The gauge group is closest to $U(1)^3$.

Degenerate limit ($\alpha \rightarrow 0$, maximum prestress). The geometric-stiffness term $(1 - \alpha)|\Delta u|^2$ dominates and contributes equal stiffness to all components on every bond. $D(k) \propto I$ be-

comes proportional to the identity at every k ; all three lateral channels are fully degenerate. The transported subspace is genuinely 3-dimensional and the adiabatic-transport object is a Wilczek–Zee connection $\mathcal{A}_\mu \in \mathfrak{u}(3)$ (Equation (7)). At the project’s default operating point $\alpha = 0.2$ the lattice sits close to this degenerate limit.

Role of microscopic anisotropy. The $U(3)$ gauge group on a 3-dimensional complex envelope is a generic representation-theoretic fact about $\mathbb{C}^3$. What the cubic anisotropy provides is the *physically meaningful decomposition*: a preferred 3-dimensional subspace (the lateral triplet, distinct from X^4), a preferred axis-aligned basis within it (which makes the $SU(3)$ traceless generators carry operational meaning as “color” labels on solitons), and a natural identification of the $U(1)$ trace with the electromagnetism-like sector. Without cubic anisotropy, no preferred basis exists and the $SU(3)$ generators would be arbitrary linear combinations with no operational handle. That anisotropy can carry genuine non-Abelian gauge structure in real space is established by precedent: Chen et al. [2] demonstrate a synthetic $SU(2)$ gauge field for optical waves from the off-diagonal permittivity of an anisotropic medium.

4.2. $U(3) \simeq U(1) \times SU(3)$ decomposition

Decomposing $\mathcal{A}_\mu \in \mathfrak{u}(3)$ into trace and traceless parts:

$$\mathcal{A}_\mu = \underbrace{\frac{1}{3}\text{tr}(\mathcal{A}_\mu)}_{U(1) \text{ part}} I_3 + \underbrace{\sum_{a=1}^8 A_\mu^a T^a}_{SU(3) \text{ part}}, \quad (12)$$

where T^a are the eight traceless $\mathfrak{su}(3)$ generators. The $U(1)$ part behaves electromagnetism-like; the traceless part carries non-Abelian self-couplings through the commutator term in $\mathcal{F}_{\mu\nu}$.

EM as the symmetric (trace) direction. The $U(1)$ /trace component of Ψ is the projection onto $(1, 1, 1)/\sqrt{3}$:

$$\Psi_{\text{EM}} = \frac{1}{\sqrt{3}}(\psi_x + \psi_y + \psi_z) \frac{(1, 1, 1)^\top}{\sqrt{3}}. \quad (13)$$

A pure-EM excitation displaces all three lateral components in phase with equal weight; the traceless complement ($SU(3)$) encodes differential phase and amplitude between components. A single-component displacement (e.g. $\psi_x \neq 0, \psi_y = \psi_z = 0$) projects as $\frac{1}{3}$ into the $U(1)$ trace and $\frac{2}{3}$ into the $SU(3)$ -traceless sector; it is not a pure-EM excitation.

Charge-triality locking. Because $U(3) = (U(1) \times SU(3))/\mathbb{Z}_3$ — the center $\mathbb{Z}_3 \subset SU(3)$ identified with the cube-roots of unity in $U(1)$ — single-valuedness forces the trace charge q and the $SU(3)$ triality $t \in \{0, 1, 2\}$ to satisfy $q \equiv -t \pmod{3}$, the standard relation for $U(N) = (U(1) \times SU(N))/\mathbb{Z}_N$ [3,4]. A color triplet therefore carries fractional ($\frac{1}{3}$ -integer) trace charge while a color singlet carries integer charge — charge fractionality is locked to color representation without reference to dynamics.

Energetic sector coupling. The central-force anharmonicity (quartic coefficient $\propto k_s \alpha / a$, Equation (20) of Paper I) couples the two sectors at quartic order through a vertex $\propto \alpha |\Psi_{\text{tr}}|^2 |\Psi_{\perp}|^2$, which vanishes identically as $\alpha \rightarrow 0$. The prestress parameter that splits the two sectors’ spectral scales is therefore also the parameter that couples them energetically.

4.3. Topological structure of the two sectors

The $U(1)$ and $SU(3)$ factors carry sharply different topology [5–7]:

- $\pi_1(U(1)) = \mathbb{Z}$ admits line vortex defects (codimension-2 in 3D), with phase winding by 2π around the core [8].
- $\pi_1(SU(3)) = 0$ forbids vortices in the $SU(3)$ sector.
- $\pi_3(SU(3)) = \mathbb{Z}$ admits Skyrmion/instanton-type textures (codimension-3 worldlines in 4D) [9,10].

The two sectors support topologically *different* classes of localized objects: codimension-2 vortex worldsheets in the EM/ $U(1)$ sector versus codimension-3 texture worldlines in the color/ $SU(3)$ sector.

However, because the full vacuum manifold is $U(3)$ — not just $U(1)$ — the $U(1)$ vortex is a *semilocal* vortex [11,12]: π_1 of the full $U(3)$ group is $\mathbb{Z}$ (the $SU(3)$ factor is simply connected), so the vortex is topologically protected within the $U(1)$ factor but not against perturbations that mix in the $SU(3)$ sector. Stability is therefore dynamical rather than purely topological, and may depend on the amplitude regime and the ratio of core energy to orientational moduli [13,14].

4.4. Kinematic confinement of color

A free linear triplet wavepacket propagating on the cubic lattice must satisfy two requirements:

Coherence. The three components stay phase-aligned over long range. This requires the three branch frequencies to be degenerate: $g(\hat{k}) = 0$, where $g(\hat{k}) = \sqrt{\sum_a (h_a - \bar{h})^2} / \bar{h}$ measures the spread of the lateral branch frequencies in direction $\hat{k}$. On the 6-neighbor cubic lattice, $g(\hat{k}) = 0$ holds *only* along the body diagonal $\hat{k} \parallel [111]$.

Color-activity. Operationally meaningful $SU(3)$ -traceless holonomy — a finite fibre-internal rotation distinguishing the three axis components — requires the branches to be *non-degenerate*: $g(\hat{k}) \neq 0$, i.e. propagation *off* $[111]$. Exactly on $[111]$ the degenerate triplet has no preferred internal basis and the color holonomy is undefined.

No linear propagation direction satisfies both simultaneously. Off $[111]$ the unequal branch speeds dephase the triplet over long range; on $[111]$ there is no resolvable color. Consequently:

Color charge cannot be carried to infinity by a free linear excitation. It has support only in a nonlinear, phase-locked bound state, where the three components are held together by the geometric coupling of Paper I rather than by spectral degeneracy.

We call this *kinematic confinement of color*. It is a statement about asymptotic states (no colored free particle), not a dynamical derivation of a linear inter-quark potential, string tension, or area law.

Falsifiable handle. The off- $[111]$ fibre-holonomy ratio scales with prestress as

$$\frac{R(\alpha_2)}{R(\alpha_1)} = \frac{(3 - 2\alpha_2)/\alpha_2}{(3 - 2\alpha_1)/\alpha_1}, \quad (14)$$

giving 0.3077 for $(\alpha_1, \alpha_2) = (0.2, 0.5)$. A deviation above $\sim 10\%$ would falsify the spectral-susceptibility factorization underlying it.

4.5. Numerical targets

The theory contains two distinct particle sectors with different topology and different numerical questions.

Target 1: $U(1)$ carrier-phase vortex (electron-like sector). The $U(1)$ trace sector supports line vortices. A preliminary rotating-frame-periodic JFNK relaxation of a pure-trace Y_1^1

vortex seed on a 112^3 lattice did not converge: the core drifted ≈ 19 lattice units off the symmetry axis, and the measured winding changed sign relative to the seed. This is the expected signature of semilocal orientational moduli with a dynamical $SU(3)$ sector. A converged eigenstate has not been obtained; branch-selection machinery for stabilizing the semilocal vortex remains future work.

Target 2: $\pi_3(SU(3))$ texture (baryon-like sector). The cubic substrate's axis-aligned triplet and $\pi_3(SU(3)) = \mathbb{Z}$ admit topologically stable Skyrmion-type textures. The canonical ansatz is the *hedgehog*

$$\xi^i(\mathbf{x}) = f(r) \hat{x}^i, \quad (15)$$

with $J = 0$, $L = 1$: internal index locked to spatial direction, invariant only under the combined diagonal $SO(3)$. The entire nontrivial structure lives in the $SU(3)$ traceless sector; the $U(1)$ trace averages to zero on the sphere, giving a trace-neutral far field (neutron-like). Topological stabilization ($B = 1$ winding) is achieved by extending the hedgehog into a Skyrme-twisted configuration. A nonzero $L = 0$ scalar admixture shifts the $U(1)$ far field to proton-like.

4.6. Color bridge open problems

1. **Verify holonomy ratio (14) numerically.** Compute the off-[111] fibre-holonomy at $\alpha = 0.2$ and $\alpha = 0.5$ and check the predicted ratio 0.3077.
2. **Converged $U(1)$ vortex eigenstate.** Develop branch-selection machinery to stabilize the semilocal vortex against orientational drift and obtain a converged periodic-BC eigenstate.
3. **Baryon/hedgehog stability test.** Construct the hedgehog ansatz on the cubic lattice, evolve under periodic-BC dynamics, and determine whether a stable, localized π_3 -winding mode exists.
4. **Dynamical co-localization of $U(1)$ and $SU(3)$ sectors.** Determine whether the quartic coupling $\propto \alpha |\Psi_{\text{tr}}|^2 |\Psi_{\perp}|^2$ is sufficient to dynamically bind the charge to the color core in a baryon-like state.

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Appendix A. Symbol dictionary

$\Omega \subset \mathbb{R}^3$	Intrinsic/material domain of the brane (coordinates $x = (x^1, x^2, x^3)$).
$\mathbf{X}(x, t) \in \mathbb{R}^4$	Embedding field of the brane into ambient 4D Euclidean space.
t	Slicing parameter along the inside observer's timelike direction ($t := x^4$); not an external/ambient time (the ambient is symmetric 4D Euclidean, Section 3.1).
g_{ij}	Induced metric: $g_{ij} = \partial_i \mathbf{X} \cdot \partial_j \mathbf{X}$.
g_{ij}^0	Isotropic reference metric: $g_{ij}^0 = \ell_0^2 \delta_{ij}$ (stress-free scale ℓ_0).
h_*	Held ground-state scale (e.g. imposed by boundary conditions).
α	Dimensionless pre-stress parameter: $\alpha = \ell_0/h_* \in (0, 1]$.
$\widehat{g}$	Mixed relative metric: $\widehat{g} = (g^0)^{-1}g$.
E	Green–Lagrange strain: $E = \frac{1}{2}(\widehat{g} - I)$.
λ, μ	Effective Lamé parameters derived from the spring constitutive law: $\mu(\alpha) = (1 - \alpha)k_s/a$, $\lambda = 0$ on the 6-neighbor stencil. StVK agrees at quadratic order only; see Section 3.5.
ρ_m	Brane mass density in the Lagrangian.
$W(g; g^0)$	Isotropic stored-energy density (hyperelastic).
P_A^i	First Piola stress: $P_A^i = \partial W / \partial (\partial_i X^A)$.
$\mathbf{u}$	Displacement field relative to a background: $\mathbf{u} = \mathbf{X} - \mathbf{X}_*$.
u	Shorthand for a transverse component in a graph-type embedding, $\mathbf{X} \approx (x, u)$.
ϵ	Normalized polarization state of a narrowband multi-component wave.
a_μ	Berry (rank-1) connection: $a_\mu = i\langle u \partial_\mu u \rangle$.
$f_{\mu\nu}$	Berry curvature: $f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$.
$\mathcal{A}_\mu$	Wilczek–Zee connection on an n -dimensional polarization subspace.
$\mathcal{F}_{\mu\nu}$	Wilczek–Zee curvature: $\partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - i[\mathcal{A}_\mu, \mathcal{A}_\nu]$.
γ_Γ	Geometric phase / holonomy around loop Γ : $\gamma_\Gamma = \oint_\Gamma a_\mu dx^\mu$.
Ω_{mix}	Mode-mixing (non-adiabatic conversion) rate between nearby polarization bands.
$\Delta\omega_{\text{gap}}$	Local spectral gap to the nearest competing polarization band.
a	Lattice spacing of the cubic substrate (microphysical length scale).
$\tilde{a}$	Momentum-lattice spacing in the discretized Brillouin zone (typically $\tilde{a} \sim 2\pi/(N_s a)$).
$\mathbf{k}$	Crystal momentum / Brillouin-zone coordinate for lattice eigenmodes.
$\Psi = (\psi_x, \psi_y, \psi_z)^\top$	Axis-aligned complex narrowband triplet carrier (cubic lattice internal sector).
T^a	Generators of $\mathfrak{su}(3)$ (e.g. Gell–Mann basis) used to decompose a $U(3)$ connection.
$R_p^l \in \mathbb{R}^4$	4D node position in the explicit lattice: $p = (i, j, k)$ spacelike triple, $l \in \mathbb{Z}$ timelike index.
k_s	Central-force spring stiffness of the 6-neighbor spacelike stencil.
m	Node mass (discrete lattice; continuum mass density $\rho_m = m/a^3$).
Δt	Temporal lattice step (timelike link length in the explicit 4D lattice).
$\mathcal{N}_s$	6-neighbor axial spacelike stencil: $\{\pm\hat{e}_x, \pm\hat{e}_y, \pm\hat{e}_z\}$.
$\mathcal{N}_t$	Temporal neighbor stencil: $\{l \pm 1\}$.
c_L	Long-wavelength longitudinal wave speed: $c_L^2 = k_s a^2 / m$.
c_T	Long-wavelength transverse wave speed: $c_T^2 = (1 - \alpha)k_s a^2 / m$.

References

1. Xiao, D.; Chang, M.C.; Niu, Q. Berry Phase Effects on Electronic Properties. *Reviews of Modern Physics* **2010**, *82*, 1959–2007. <https://doi.org/10.1103/RevModPhys.82.1959>.
2. Chen, Y.; Zhang, R.Y.; Xiong, Z.; Hang, Z.H.; Li, J.; Shen, J.Q.; Chan, C.T. Non-Abelian gauge field optics. *Nature Communications* **2019**, *10*, 3125. Synthetic SU(2) gauge field from material anisotropy in homogeneous media; off-diagonal permittivity/permeability act as spin-dependent vector potentials; predicts Zitterbewegung trajectories and a non-Abelian Aharonov–Bohm effect with extractable Wilson loop, <https://doi.org/10.1038/s41467-019-10974-8>.
3. Georgi, H.; Glashow, S.L. Unity of All Elementary-Particle Forces. *Physical Review Letters* **1974**, *32*, 438–441. Georgi–Glashow SU(5): the smallest simple group embedding $SU(3) \times SU(2) \times U(1)$; one unified coupling at M_G , <https://doi.org/10.1103/PhysRevLett.32.438>.
4. Baez, J.C.; Huerta, J. The Algebra of Grand Unified Theories. *Bulletin of the American Mathematical Society* **2010**, *47*, 483–552, [arXiv:hep-th/0904.1556]. Hypercharge normalization and the $\mathbb{Z}_6$ quotient $G_{SM}/\mathbb{Z}_6 = S(U(2) \times U(3)) \subset SU(5)$ underlying charge quantization, <https://doi.org/10.1090/S0273-0979-10-01294-2>.
5. Kibble, T.W.B. Topology of Cosmic Domains and Strings. *Journal of Physics A: Mathematical and General* **1976**, *9*, 1387–1398. Homotopy classification of defects: $\pi_0 \rightarrow$ walls, $\pi_1 \rightarrow$ strings, $\pi_2 \rightarrow$ monopoles, $\pi_3 \rightarrow$ textures, <https://doi.org/10.1088/0305-4470/9/8/029>.
6. Vilenkin, A.; Shellard, E.P.S. *Cosmic Strings and Other Topological Defects*; Cambridge University Press, 1994. Standard monograph on the $\pi_n(G/H)$ defect spectrum after symmetry breaking.
7. Manton, N.; Sutcliffe, P. *Topological Solitons*; Cambridge Monographs on Mathematical Physics, Cambridge University Press, 2004. Standard reference for vortices, monopoles, skyrmions, instantons and the Atiyah–Manton construction, <https://doi.org/10.1017/CBO9780511617034>.
8. Nielsen, H.B.; Olesen, P. Vortex-Line Models for Dual Strings. *Nuclear Physics B* **1973**, *61*, 45–61. Abelian-Higgs vortex line; $\pi_1(U(1)) = \mathbb{Z}$ winding quantizes flux as $\Phi = (2\pi/e)n$ – the EM-like codim-2 line defect, [https://doi.org/10.1016/0550-3213\(73\)90350-7](https://doi.org/10.1016/0550-3213(73)90350-7).
9. Skyrme, T.H.R. A Unified Field Theory of Mesons and Baryons. *Nuclear Physics* **1962**, *31*, 556–569. Baryon as an SU(2) π_3 texture; winding number identified with baryon number, [https://doi.org/10.1016/0029-5582\(62\)90775-7](https://doi.org/10.1016/0029-5582(62)90775-7).
10. Atiyah, M.F.; Manton, N.S. Skyrmions from Instantons. *Physics Letters B* **1989**, *222*, 438–442. Builds 3D π_3 skyrmion baryons from 4D SU(N) Yang–Mills instanton holonomy, [https://doi.org/10.1016/0370-2693\(89\)90340-7](https://doi.org/10.1016/0370-2693(89)90340-7).
11. Vachaspati, T.; Achúcarro, A. Semilocal Cosmic Strings. *Physical Review D* **1991**, *44*, 3067–3071. Stable codim-2 vortex on a simply-connected (π_1 trivial) vacuum for $\beta < 1$: stability is dynamical, not topological – key precedent for an SU(3)-unfrozen electron vortex, <https://doi.org/10.1103/PhysRevD.44.3067>.
12. Achúcarro, A.; Vachaspati, T. Semilocal and Electroweak Strings. *Physics Reports* **2000**, *327*, 347–426, [arXiv:hep-ph/hep-ph/9904229]. Review; the $n = 1$ semilocal vortex carries an internal $CP^1 = SU(2)/U(1)$ orientational modulus – a U(1) line defect inside a larger non-Abelian symmetry; $\beta =$ type-I/II superconductor parameter, [https://doi.org/10.1016/S0370-1573\(99\)00103-9](https://doi.org/10.1016/S0370-1573(99)00103-9).
13. Auzzi, R.; Bolognesi, S.; Evslin, J.; Konishi, K.; Yung, A. Nonabelian Superconductors: Vortices and Confinement in $N = 2$ SQCD. *Nuclear Physics B* **2003**, *673*, 187–216, [arXiv:hep-th/hep-th/0307287]. Non-Abelian vortex with $CP^{N-1} = SU(N)/[SU(N-1) \times U(1)]$ orientational moduli from color-flavor locking, <https://doi.org/10.1016/j.nuclphysb.2003.09.029>.
14. Eto, M.; Hamada, Y.; Nitta, M.; et al. Dynamics of Global and Local Vortices with Orientational Moduli. *Journal of High Energy Physics* **2021**, *2021*, 156, [arXiv:hep-th/2012.04098]. U(1)-gauged + global SO(N) vortex with core-localized S^{N-1} orientational Nambu–Goldstone moduli, [https://doi.org/10.1007/JHEP03\(2021\)156](https://doi.org/10.1007/JHEP03(2021)156).
15. Molzberger, L. BraneSim: a deterministic brane-lattice substrate simulator. <https://github.com/LukasMolzberger/BraneSim>, 2026. Open-source substrate solver, diagnostic suite, and experiment pipeline reproducing the figures in this paper.

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